

LAPORAN JOBSHEET PEMROGRAMAN BERBASIS FRAMEWORK



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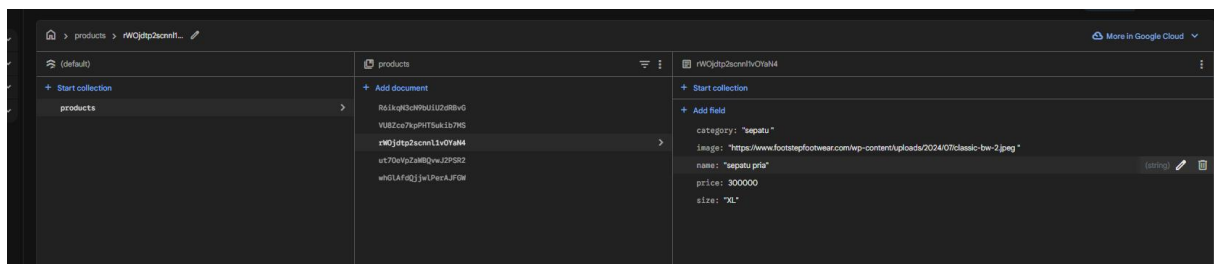
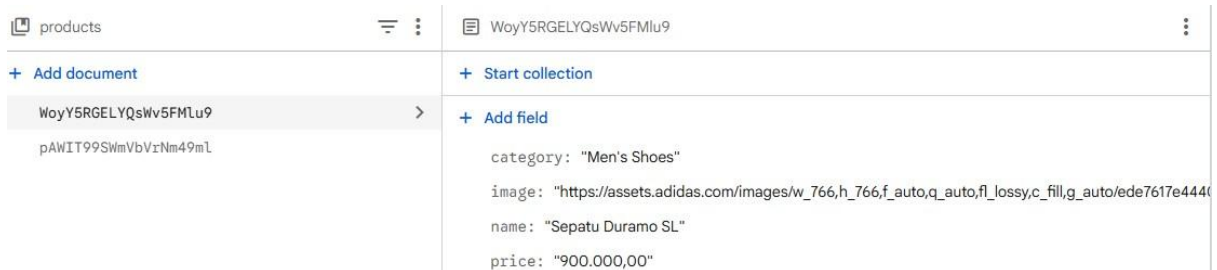
NIM: 2341720142

Kelas: TI 3E

POLITEKNIK NEGERI MALANG TAHUN 2026

C. Langkah Praktikum

Buka firebase dan rubah fieldnya serta tambahkan beberapa field seperti pada gambar dan untuk image bisa gunakan url pada image toko-toko sepatu (copy image address). Selain itu tambah juga documentnya sehingga ada 2 document



◆ Bagian 1 – Setup Data Produk

1. Siapkan project Next.js.
2. Buat endpoint API /api/products.
3. Pastikan data memiliki:
 - id
 - name
 - category
 - price
 - Image

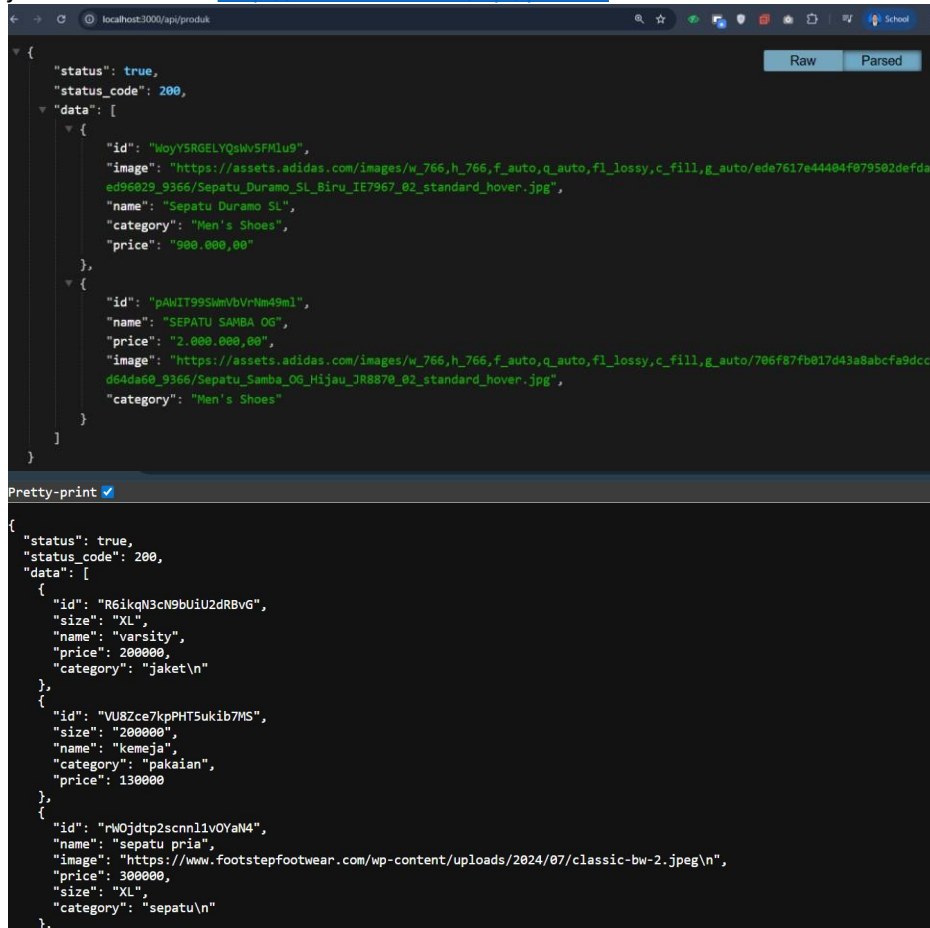
```
src > pages > produk > index.tsx > Kategori > products.map() callback
1  import { useEffect, useState } from "react";
2
3  Windsurf: Refactor | Explain
4  type ProductType = {
5    id: string;
6    name: string;
7    price: number;
8    size: string;
9    category: string;
10   image: string;
11 };
12
```

```

50 </button>
51
52 {products.map((product) => [
53   <div key={product.id}>
54     <h2>{product.name}</h2>
55     <p>Harga: {product.price}</p>
56     <p>Ukuran: {product.size}</p>
57     <p>Kategori: {product.category}</p>
58     <img src={product.image} alt={product.name} width="100" height="100" />
59   </div>
60 ])}
61 </div>
62 );
63 };
64
65 export default Kategori;
66

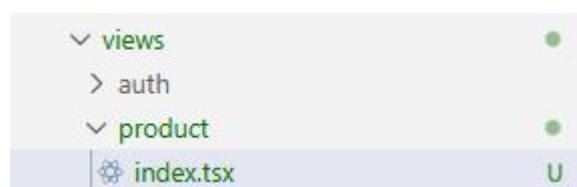
```

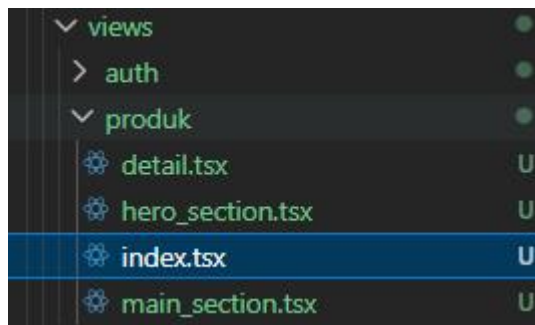
4. jalankan browser <http://localhost:3000/api/produk>



◆ Bagian 2 – Implementasi CSR dengan useEffect

1. Membuat file index.tsx pada folder views/products





2. Modifikasi index.tsx

7.praktikum-Client_side rendering > my-app > src > pages > views > product > index.tsx > ...

```

1
Windsurf: Refactor | Explain
2 type ProductType = {
3   id: string;
4   name: string;
5   price: number;
6   image: string;
7   category: string;
8 };
9
Windsurf: Refactor | Explain | Generate JSDoc | X
10 const TampilanProduk = ({ products }: { products: ProductType[] }) => {
11   return (
12     <div>
13       <h1>Daftar Produk</h1>
14       {products.map((products: ProductType) => (
15         <div key={products.id}>
16           <h2>nama :{products.name}</h2>
17           <p>Harga: {products.price}</p>
18           <img src={products.image} alt={products.name} width={200} />
19           <p>kategori: {products.category}</p>
20         </div>
21       ))}
22     </div>
23   );
24 };
25
26 export default TampilanProduk;
27

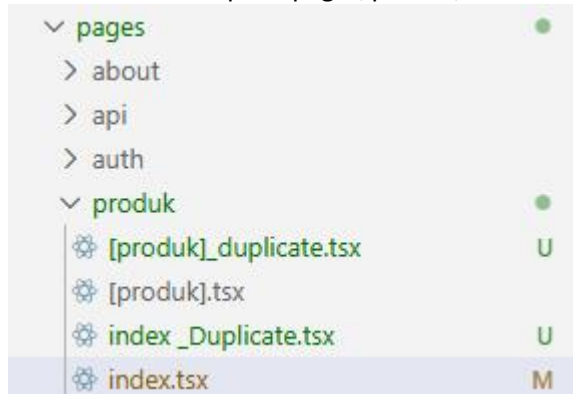
```

```

Windsurf: Refactor | Explain
1 type productType = {
2   id: string;
3   name: string;
4   price: number;
5   image: string;
6   category: string;
7 };
8
Windsurf: Refactor | Explain | Generate JSDoc | X
9 const TampilanProduk = ({products}: {products: productType[]}) => {
10   return (
11     <div>
12       {products.map((products:productType) => (
13         <div key={products.id}>
14           <h2>nama : {products.name}</h2>
15           <p>harga : {products.price}</p>
16           <img src={products.image} alt={products.name} width={200} />
17           <p>kategori : {products.category}</p>
18         </div>
19       ))}
20     </div>
21   );
22 };
23
24 export default TampilanProduk;

```

3. Buka file index.tsx pada pages/produk/



4. Modifikasi index.tsx pada pages/produk/

7.praktikum-Client_side rendering > my-app > src > pages > produk > index.tsx > kategori > useEffect() callback > then() callback

```
1 import { useRouter } from "next/router";
2 import { useEffect, useState } from "react";
3 import TampilanProduk from "../views/product";
4
5 const kategori = () => {
6   // const [isLogin, setIsLogin] = useState(false);
7   // const { push } = useRouter();
8   const [products, setProducts] = useState([]);
9   // console.log("products:", products);
10  // useEffect(() => {
11  //   if (!isLogin) {
12  //     push("/auth/login");
13  //   }
14  // }, []);
15
16  useEffect(() => {
17    fetch("/api/produk")
18      .then((response) => response.json())
19      .then((responseData) => {
20        setProducts(responseData.data);
21        // console.log("Data produk:", responseData.data);
22      })
23      .catch((error) => {
24        console.error("Error fetching produk:", error);
25      });
26  }, []);
27
28  return (
29    <div>
30      <TampilanProduk products={products} />
31    </div>
32  );
33 };
34
35 export default kategori;
```

```

import { useRouter } from "next/router";
import { useEffect, useState } from "react";
import TampilanProduk from "../views/produk";

Windsurf: Refactor | Explain | Generate JSDoc | X
const kategori = () => {
  // const [isLoggedIn, setIsLoggedIn] = useState(false);
  // const { push } = useRouter();
  const [products, setProducts] = useState([]);
  // console.log("products:", products);

  // useEffect(() => {
  //   if (!isLoggedIn) {
  //     push("/auth/login");
  //   }
  // }, []);

  useEffect(() => {
    fetch("/api/produk")
      .then((response) => response.json())
      .then((responsedata) => {
        setProducts(responsedata.data);
        // console.log("Data produk:", responsedata.data);
      })
      .catch((error) => {
        console.error("Error fetching produk:", error);
      });
  }, []);

  return (
    <div>
      <TampilanProduk products={products} />
    </div>
  );
};

export default kategori;

```

5. Jalankan browser <http://localhost:3000/produk>

Navbar

Daftar Produk

nama :Sepatu Duramo SL

Harga: 900.000,00



Harga: Men's Shoes

nama :SEPATU SAMBA OG

Harga: 2.000.000,00



Harga: Men's Shoes



Navbar

nama : varsity

harga : 200000

 varsity

kategori : jaket

nama : kemeja

harga : 130000

 kemeja

kategori : pakaian

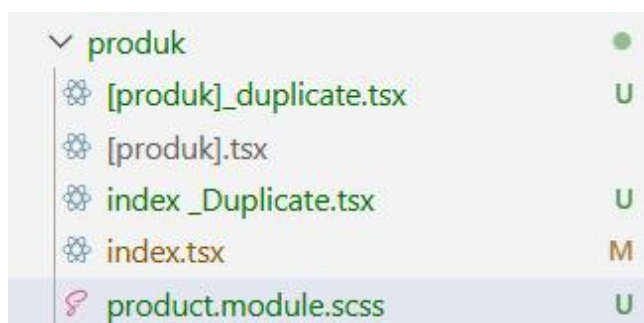
nama : sepatu pria

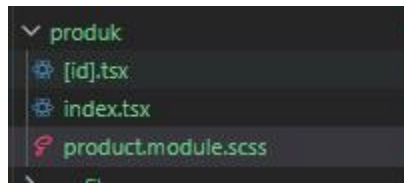
harga : 300000



kategori : sepatu

6. Pada folder produk buat file produk.modules.scss





7. Modifikasi produk.modules.scss

```
7.praktikum-Client_side rendering > my-app > src > pages > produk > product.module.scss > .produk > &_content > &_item
1  .produk {
2    width: 100%;
3    padding: 0 5%;
4    &_title {
5      text-align: center;
6      font-size: 32px;
7      font-weight: bold;
8      margin-bottom: 16px;
9    }
10   &_content {
11     display: flex;
12     flex-wrap: wrap;
13     gap: 20px;
14     &_item {
15       &_image {
16         width: 100%;
17         height: 200px;
18         object-fit: cover;
19         border-radius: 4px;
20         margin-bottom: 12px;
21       }
22       &_name {
23         font-size: 18px;
24         font-weight: 600;
25         margin-bottom: 8px;
26       }
27       &_price {
28         font-size: 16px;
29         color: #ff5722;
30         font-weight: bold;
31       }
32       &_category {
33         font-size: 14px;
34         color: #888;
35         margin-bottom: 8px;
36       }
37     }
38   }
39 }
40
```

```

src > pages > produk > product.module.scss >
1  .produk {
2    width: 100%;
3    padding: 0 5%;
4
5    &__title {
6      text-align: center;
7      font-size: 32px;
8      font-weight: bold;
9      margin-bottom: 16px;
10   }
11
12   &__content {
13     display: flex;
14     flex-wrap: wrap;
15     gap: 20px;
16
17     &__item {
18       &__image {
19         width: 100%;
20         height: 200px;
21         object-fit: cover;
22         border-radius: 4px;
23         margin-bottom: 12px;
24       }
25
26       &__name {
27         font-size: 18px;
28         font-weight: 600;
29         margin-bottom: 8px;
30       }
31
32       &__price {
33         font-size: 16px;
34         color: #ff5722;
35         font-weight: bold;
36       }
37
38       &__category {
39         font-size: 14px;
40         color: #888;
41         margin-bottom: 8px;
42       }
43     }
44   }
45 }

```

8. Modifikasi Pada file index.tsx pada folder pages/views/product

```

7.praktikum-Client_side rendering > my-app > src > pages > views > product > index.tsx > ...
1  import styles from "../../produk/product.module.scss";
2
3
4  Windsurf: Refactor | Explain
5  type ProductType = {
6    id: string;
7    name: string;
8    price: number;
9    image: string;
10   category: string;
11 };
12
13 Windsurf: Refactor | Explain | Generate JSDoc | X
14 const TampilanProduk = ({ products }: { products: ProductType[] }) => {
15   return (
16     <div className={styles.produk}>
17       <h1 className={styles.produk__title}>Daftar Produk</h1>
18       <div className={styles.produk__content}>
19         {products.map((products: ProductType) => (
20           <div key={products.id} className={styles.produk__content__item}>
21             <div className={styles.produk__content__item__image}>
22               <img src={products.image} alt={products.name} width={200} />
23             </div>
24             <h4 className={styles.produk__content__item__name}>
25               {products.name}
26             </h4>
27             <p className={styles.produk__content__item__category}>
28               {products.category}
29             </p>
30             <p className={styles.produk__content__item__price}>
31               Rp {products.price.toLocaleString()}
32             </p>
33           </div>
34         ))}
35       </div>
36     </div>
37   );
38 };
39
40 export default TampilanProduk;

```

```

src > pages > views > produk > index.tsx > ...
1  import styles from "../../produk/product.module.scss";
2
Windsurf: Refactor | Explain
3  type ProductType = {
4    id: string;
5    name: string;
6    price: number;
7    image: string;
8    category: string;
9  };
10
Windsurf: Refactor | Explain | Generate JSDoc | X
11  const TampilanProduk = ({ products }: { products: ProductType[] }) => {
12    return (
13      <div className={styles.produk}>
14        <h1 className={styles.produk__title}>Daftar Produk</h1>
15        <div className={styles.produk__content}>
16          {products.map((product: ProductType) => (
17            <div key={product.id} className={styles.produk__content__item}>
18              <div className={styles.produk__content__item__image}>
19                <img src={product.image} alt={product.name} width={200} />
20              </div>
21              <h4 className={styles.produk__content__item__name}>
22                {product.name}
23              </h4>
24              <p className={styles.produk__content__item__category}>
25                {product.category}
26              </p>
27              <p className={styles.produk__content__item__price}>
28                Rp {product.price.toLocaleString()}
29              </p>
30            </div>
31          ))}
32        </div>
33      </div>
34    );
35  };
36
37  export default TampilanProduk;

```

9. Jalankan Browser



Daftar Produk



Sepatu Duramo SL

Men's Shoes

Rp 900.000,00

SEPATU SAMBA OG

Men's Shoes

Rp 2.000.000,00



Daftar Produk



varsity



kemeja



celana



baju polo



varsity

jaket

Rp 200,000

kemeja

pakaian

Rp 130,000

sepatu pria

sepatu

Rp 300,000

celana

celana

Rp 150,000

baju polo

pakaian

Rp 100,000

◆ Bagian 3 – Implementasi Skeleton Loading

- Modifikasi file index.tsx pada folder views/product/index.tsx

```
7.praktikum-Client_side rendering > my-app > src > pages > views > product > index.tsx > ...
1  import styles from "../../produk/product.module.scss";
   Windsurf: Refactor | Explain
2  type ProductType = {
3    id: string;
4    name: string;
5    price: number;
6    image: string;
7    category: string;
8  };
9
   Windsurf: Refactor | Explain | Generate JSDoc | X
10 const TampilanProduk = ({ products }: { products: ProductType[] }) => {
11   return (
12     <div className={styles.produk}>
13       <h1 className={styles.produk_title}>Daftar Produk</h1>
14       <div className={styles.produk_content}>
15         { /* {products.map((products: ProductType) => (
16           <div key={products.id} className={styles.produk_content_item}>
17             <div className={styles.produk_content_item_image}>
18               <img src={products.image} alt={products.name} width={200} />
19             </div>
20             <h4 className={styles.produk_content_item_name}>
21               {products.name}
22             </h4>
23             <p className={styles.produk_content_item_category}>
24               {products.category}
25             </p>
26             <p className={styles.produk_content_item_price}>
27               Rp {products.price.toLocaleString()}
28             </p>
29           </div>
30         )} */ }
31         <div className={styles.produk_content_skeleton}>
32           <div className={styles.produk_content_skeleton_image}></div>
33           <div className={styles.produk_content_skeleton_name}></div>
34           <div className={styles.produk_content_skeleton_category}></div>
35           <div className={styles.produk_content_skeleton_price}></div>
36         </div>
37       </div>
38     </div>
39   );
40 };
41 export default TampilanProduk;
```

```

src > pages > views > produk > index.jsx > default
1  import styles from "../../produk/product.module.scss";
2
Windsurf: Refactor | Explain
3  type ProductType = {
4      id: string;
5      name: string;
6      price: number;
7      image: string;
8      category: string;
9  };
10
Windsurf: Refactor | Explain | Generate JSDoc | X
11 const TampilanProduk = ({ products }: { products: ProductType[] }) => {
12     return (
13         <div className={styles.produk}>
14             <h1 className={styles.produk__title}>Daftar Produk</h1>
15             <div className={styles.produk__content}>
16                 {/* {products.map((products: ProductType) => {
17                     <div key={products.id} className={styles.produk__content__item}>
18                         <div className={styles.produk__content__item__image}>
19                             <img src={products.image} alt={products.name} width={200} />
20                         </div>
21                         <h4 className={styles.produk__content__item__name}>
22                             {products.name}
23                         </h4>
24                         <p className={styles.produk__content__item__category}>
25                             {products.category}
26                         </p>
27                         <p className={styles.produk__content__item__price}>
28                             Rp {products.price.toLocaleString()}
29                         </p>
30                     </div>
31                 })} */
32                 <div className={styles.produk__content__skeleton}>
33                     <div className={styles.produk__content__skeleton__image}></div>
34                     <div className={styles.produk__content__skeleton__name}></div>
35                     <div className={styles.produk__content__skeleton__category}></div>
36                     <div className={styles.produk__content__skeleton__price}></div>
37                 </div>
38             </div>
39         </div>
40     );
41 };
42
43 export default TampilanProduk;

```

- Modifikasi file product.module.scss

```
1  .produkt {
2    width: 100%;
3    padding: 0 5%;
4    &__title {
5      text-align: center;
6      font-size: 32px;
7      font-weight: bold;
8      margin-bottom: 16px;
9    }
10   &__content {
11     display: flex;
12     flex-wrap: wrap;
13     gap: 20px;
14     &__skeleton {
15       width: 200px;
16       padding: 16px;
17       border: 1px solid #eee;
18       border-radius: 8px;
19       box-shadow: 0 2px 4px rgba(0, 0, 0, 0.1);
20       display: flex;
21       flex-direction: column;
22       align-items: center;
23       animation: identifier 1.5s infinite ease-in-out;
24       &__image {
25         width: 100%;
26         height: 200px;
27         background-color: #e0e0e0;
28         border-radius: 4px;
29         margin-bottom: 12px;
30       }
31       &__name {
32         width: 80%;
33         height: 20px;
34         background-color: #e0e0e0;
35         border-radius: 4px;
36         margin-bottom: 8px;
37       }
38       &__category {
39         width: 60%;
40         height: 16px;
41         background-color: #e0e0e0;
42         border-radius: 4px;
43         margin-bottom: 8px;
44       }
45       &__price {
46         width: 40%;
47         height: 18px;
48         background-color: #e0e0e0;
49         border-radius: 4px;
50       }
51     }
52   }
53   &__item {
54     &__image {
55       width: 100%;
56       height: 200px;
57       object-fit: cover;
58       border-radius: 4px;
59       margin-bottom: 12px;
60     }
61     &__name {
62       font-size: 18px;
63       font-weight: bold;
64       margin-bottom: 8px;
65     }
66     &__price {
67       font-size: 16px;
68       color: #ff5722;
69       font-weight: bold;
70     }
71     &__category {
72       font-size: 14px;
73       color: #888;
74       margin-bottom: 8px;
75     }
76   }
77 }
78 }
79
80 @keyframes identifier {
81   0% {
82     opacity: 1;
83   }
84   50% {
85     opacity: 0;
86   }
87   100% {
88     opacity: 1;
89   }
90 }
```



```

.produk {
  width: 100%;
  padding: 0 5%;

  &__title {
    text-align: center;
    font-size: 32px;
    font-weight: bold;
    margin-bottom: 16px;
  }

  &__content {
    display: flex;
    flex-wrap: wrap;
    gap: 20px;

    &__skeleton {
      width: 200px;
      padding: 16px;
      border: 1px solid #eee;
      border-radius: 8px;
      box-shadow: 0 2px 4px rgba(0, 0, 0, 0.1);
      display: flex;
      flex-direction: column;
      align-items: center;
      animation: identifier 1.5s infinite ease-in-out;

      &__image {
        width: 100%;
        height: 200px;
        background-color: #e0e0e0;
        border-radius: 4px;
        margin-bottom: 12px;
      }

      &__name {
        width: 80%;
        height: 20px;
        background-color: #e0e0e0;
        border-radius: 4px;
        margin-bottom: 8px;
      }
    }
  }
}

```

```

    &__category {
      width: 60%;
      height: 16px;
      background-color: #e0e0e0;
      border-radius: 4px;
      margin-bottom: 8px;
    }

    &__price {
      width: 40%;
      height: 18px;
      background-color: #e0e0e0;
      border-radius: 4px;
    }
  }

  &__item {
    &__image {
      width: 100%;
      height: 200px;
      object-fit: cover;
      border-radius: 4px;
      margin-bottom: 12px;
    }

    &__name {
      font-size: 18px;
      font-weight: 600;
      margin-bottom: 8px;
    }

    &__price {
      font-size: 16px;
      color: #ff5722;
      font-weight: bold;
    }

    &__category {
      font-size: 14px;
      color: #888;
      margin-bottom: 8px;
    }
  }
}
}

```

```

@keyframes identifier {
  0% {
    opacity: 1;
  }
  50% {
    opacity: 0;
  }
  100% {
    opacity: 1;
  }
}

```

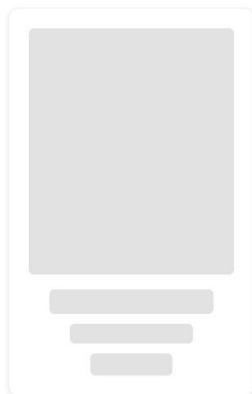
- Jalankan browser maka akan muncul skeleton yang terdapat animasi berkedip

Navbar

Daftar Produk

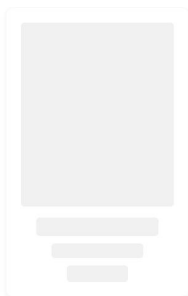
Navbar

Daftar Produk



Navbar

Daftar Produk



footer

- Modifikasi pada index.tsx pada folder views/product/index.tsx

7.praktikum-Client_side rendering > my-app > src > pages > views > product > index.tsx > ...

```
1  import styles from "../../produk/product.module.scss";
   Windsurf: Refactor | Explain
2  type ProductType = {
3    id: string;
4    name: string;
5    price: number;
6    image: string;
7    category: string;
8  };
9
   Windsurf: Refactor | Explain | Generate JSDoc | X
10 const TampilanProduk = ({ products }: { products: ProductType[] }) => {
11   return (
12     <div className={styles.produk}>
13       <h1 className={styles.produk_title}>Daftar Produk</h1>
14       <div className={styles.produk_content}>
15         {products.length > 0 ? (
16           <>
17             {products.map((products: ProductType) => (
18               <div key={products.id} className={styles.produk_content_item}>
19                 <div className={styles.produk_content_item_image}>
20                   <img src={products.image} alt={products.name} width={200} />
21                 </div>
22                 <h4 className={styles.produk_content_item_name}>
23                   {products.name}
24                 </h4>
25                 <p className={styles.produk_content_item_category}>
26                   {products.category}
27                 </p>
28                 <p className={styles.produk_content_item_price}>
29                   Rp {products.price.toLocaleString()}
30                 </p>
31               </div>
32             ))}
33           </>
34         ) : (
35           <div className={styles.produk_content_skeleton}>
36             <div className={styles.produk_content_skeleton_image}></div>
37             <div className={styles.produk_content_skeleton_name}></div>
38             <div className={styles.produk_content_skeleton_category}></div>
39             <div className={styles.produk_content_skeleton_price}></div>
40           </div>
41         )}
42       </div>
43     </div>
44   );
45 };
46 export default TampilanProduk;
```

```

1 import styles from "../../produk/product.module.scss";
2
3 type ProductType = {
4   id: string;
5   name: string;
6   price: number;
7   image: string;
8   category: string;
9 };
10
11 const TampilanProduk = ({ products }: { products: ProductType[] }) => {
12   return (
13     <div className={styles.produk}>
14       <h1 className={styles.produk__title}>Daftar Produk</h1>
15       <div className={styles.produk__content}>
16         {products.length > 0 ? (
17           <>
18             {products.map((products: ProductType) => (
19               <div key={products.id} className={styles.produk__content__item}>
20                 <div className={styles.produk__content__item__image}>
21                   <img src={products.image} alt={products.name} width={200} />
22                 </div>
23                 <h4 className={styles.produk__content__item__name}>
24                   {products.name}
25                 </h4>
26                 <p className={styles.produk__content__item__category}>
27                   {products.category}
28                 </p>
29                 <p className={styles.produk__content__item__price}>
30                   Rp {products.price.toLocaleString()}
31                 </p>
32               </div>
33             ))}
34           </>
35         ) : (
36           <div className={styles.produk__content__skeleton}>
37             <div className={styles.produk__content__skeleton__image}></div>
38             <div className={styles.produk__content__skeleton__name}></div>
39             <div className={styles.produk__content__skeleton__category}></div>
40             <div className={styles.produk__content__skeleton__price}></div>
41           </div>
42         )}
43       </div>
44     </div>
45   );
46 };
47
48 export default TampilanProduk;

```

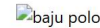
- Jalankan browser
 - Jika dijalankan akan muncul skeletonya terlebih dahulu setelah itu muncul gambar dan informasinya

Daftar Produk

 varsity

 kemeja

 celana

 baju polo


varsity

jaket

Rp 200,000

kemeja

pakaian

Rp 130,000

sepatu pria

sepatu

Rp 300,000

celana

celana

Rp 150,000

baju polo

pakaian

Rp 100,000

voter

◆ Bagian 5 – Implementasi SWR

<https://swr.vercel.app/>

1. Install SWR

```
PS D:\1.MateriKuliah\2026-ganjil\PemogramanFramework\Code\7.praktikum-Client_side rendering\my-app> npm install swr
added 3 packages, and audited 429 packages in 2s
142 packages are looking for funding
run `npm fund` for details
found 0 vulnerabilities
```

```
D:\tugas\semester 6\framework\js8\code\code\my-app>npm install swr
added 3 packages, and audited 471 packages in 59s
```

2. Buka dan modifikasi file index.tsx pada folder pages/product/

```
7.praktikum-Client_side rendering > my-app > src > pages > produk > index.tsx > kategori
1  import { useRouter } from "next/router";
2  import { useEffect, useState } from "react";
3  import TampilanProduk from "../views/product";
4  import useSWR from "swr";
5
6  const fetcher = (url: string) => fetch(url).then((res) => res.json());
Windsurf: Refactor | Explain | Generate JSDoc | X
7  const kategori = () => {
8  > // const [isLoggedIn, setIsLoggedIn] = useState(false); ...
10 // const [products, setProducts] = useState([]);
11 > // console.log("products:", products); ...
29
30 const { data, error, isLoading } = useSWR("/api/produk", fetcher);
31 //cek apakah data, error, dan isLoading sudah benar
32 // console.log("data:", data);
33 // console.log("error:", error);
34 // console.log("isLoading:", isLoading);
35
36
37 return (
38 <div>
39 <TampilanProduk products={isLoading ? [] : data.data} />
40 </div>
41 );
42 };
43
44 export default kategori;
45
```

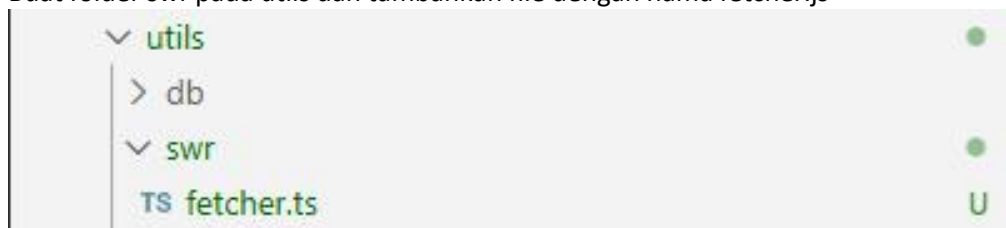
```

src / pages / produk > @ index.tsx > ...
1 > // import { useEffect, useState } from "react";...
66 import { useRouter } from "next/router";
67 import { useEffect, useState } from "react";
68 import TampilanProduk from "../views/produk";
69 import useSWR from "swr";
70
71 const fetcher = (url: string) => fetch(url).then((res) => res.json());
72
73 Windsurf Refactor | Explain | Generate JSDoc | X
74 const kategori = () => {
75   // const [islogin, setIslogin] = useState(false);
76   // const { push } = useRouter();
77   const [products, setProducts] = useState([]);
78   // console.log("products:", products);
79
80   // useEffect(() => {
81   //   if (!islogin) {
82   //     push("/auth/login");
83   //   }
84   // }, []);
85
86   const { data, error, isLoading } = useSWR("/api/produk", fetcher);
87
88   // useEffect(() => {
89   //   fetch("/api/produk")
90   //     .then((response) => response.json())
91   //     .then((responseData) => {
92   //       setProducts(responseData.data);
93   //       // console.log("Data produk:", responseData.data);
94   //     })
95   //     .catch((error) => {
96   //       console.error("Error fetching produk:", error);
97   //     });
98   // }, []);
99
100   return (
101     <div>
102       <TampilanProduk products={isLoading ? [] : data?.data} />
103     </div>
104   );
105
106   export default kategori;
107

```

3. Agar terlihat lebih rapi

- Buat folder swr pada utils dan tambahkan file dengan nama fetcher.js



- Modifikasi file fetcher.ts

7.praktikum-Client_side rendering > my-app > src > pages > utils > swr > TS fetcher.ts > default

```
1
2 const fetcher = (url: string) => fetch(url).then((res) => res.json());
3
4 export default fetcher;
```

```
src > pages > utils > swr > TS fetcher.ts > default
1
2 const fetcher = (url: string) => fetch(url).then((res) => res.json());
3
4 export default fetcher;
```

- Modifikasi file index.tsx pada folder pages/produk

7.praktikum-Client_side rendering > my-app > src > pages > produk > index.tsx > ...

```
1 import { useRouter } from "next/router";
2 import { useEffect, useState } from "react";
3 import TampilanProduk from "../views/product";
4 import useSWR from "swr";
5 import fetcher from "../utils/swr/fetcher";
6
7
8
9 // const fetcher = (url: string) => fetch(url).then((res) => res.json());
10 Windsurf: Refactor | Explain | X
11 const kategori = () => {
12   // const [isLogin, setIsLogin] = useState(false);
13   // const { push } = useRouter();
14   const [products, setProducts] = useState([]);
15   // console.log("products:", products); ...
16
17
18
19
20
21
22
23 const { data, error, isLoading } = useSWR("/api/produk", fetcher);
24 //cek apakah data, error, dan isLoading sudah benar ...
25
26
27
28
29
30
31
32
33
34
35
36
37
38
39
40 return (
41   <div>
42     <TampilanProduk products={isLoading ? [] : data.data} />
43   </div>
44 );
45 };
46
47 export default kategori;
48
```

```

1 > // import { useEffect, useState } from "react";...
66 import { useRouter } from "next/router";
67 import { useEffect, useState } from "react";
68 import TampilanProduk from "../views/produk";
69 import useSWR from "swr";
70 import fetcher from '../utils/swr/fetxher';
71
72 // const fetcher = (url: string) => fetch(url).then((res) => res.json());
73
Windsurf: Refactor | Explain | Generate JSDoc | X
74 const kategori = () => {
75   // const [isLoggedIn, setIsLogin] = useState(false);
76   // const { push } = useRouter();
77   const [products, setProducts] = useState([]);
78   // console.log("products:", products);
79
80   // useEffect(() => {
81   //   if (!isLoggedIn) {
82   //     push("/auth/login");
83   //   }
84   // }, []);
85
86   const { data, error, isLoading } = useSWR("/api/produk", fetcher);
87
88   // useEffect(() => {
89   //   fetch("/api/produk")
90   //     .then((response) => response.json())
91   //     .then((responseData) => {
92   //       setProducts(responseData.data);
93   //       // console.log("Data produk:", responseData.data);
94   //     })
95   //     .catch((error) => {
96   //       console.error("Error fetching produk:", error);
97   //     });
98   // }, []);
99
100   return (
101     <div>
102       <TampilanProduk products={isLoading ? [] : data.data} />
103     </div>
104   );
105 };
106
107 export default kategori;
108

```

Bandingkan:

- useEffect manual
- SWR

Note : untuk client side rendering dapat menggunakan 2 cara yaitu menggunakan manual (useEffect) atau menggunakan SWR

D. Tugas Praktikum

Tugas Individu

1. Jelaskan perbedaan:
 - Client Side Rendering
 - Server Side Rendering

- Static Site Generation

Jawab :

- Client Side Rendering (CSR) adalah metode di mana halaman web dirender di browser menggunakan JavaScript setelah file dikirim dari server. Server hanya mengirimkan HTML dasar dan script, lalu browser yang mengambil data dan membangun tampilan. Metode ini cocok untuk aplikasi interaktif, tetapi kurang optimal untuk SEO.
- Server Side Rendering (SSR) adalah metode di mana halaman dirender terlebih dahulu di server setiap kali ada permintaan dari pengguna. Server mengirimkan HTML yang sudah lengkap sehingga halaman bisa langsung ditampilkan dan lebih baik untuk SEO, namun beban server menjadi lebih besar.
- Static Site Generation (SSG) adalah metode di mana halaman dibuat saat proses build dan disimpan sebagai file HTML statis. Saat diakses, server langsung mengirimkan file yang sudah jadi. Metode ini sangat cepat dan baik untuk SEO, tetapi tidak cocok untuk data yang sering berubah.

2. Buat halaman produk dengan:

- Skeleton loading
- Animasi

index.tsx - my-app - Visual Studio Code

```
src > pages > views > produk > index.tsx > ProductType
1  import styles from "../../produk/product.module.scss";
2
3  Windsurf: Refactor | Explain
4  type ProductType = {
5      id: string;
6      name: string;
7      price: number;
8      image: string;
9      category: string;
10 };
11
12 Windsurf: Refactor | Explain | Generate JSDoc | X
13 const TampilanProduk = ({ products }: { products: ProductType[] }) => {
14     return (
15         <div className={styles.produk}>
16             <h1 className={styles.produk_title}>Daftar Produk</h1>
17             <div className={styles.produk_content}>
18                 {products.length > 0 ? (
19                     <>
20                         {products.map((products: ProductType) => (
21                             <div key={products.id} className={styles.produk_content_item}>
22                                 <div className={styles.produk_content_item_image}>
23                                     <img src={products.image} alt={products.name} width={200} />
24                                 </div>
25                                 <h4 className={styles.produk_content_item_name}>
26                                     {products.name}
27                                 </h4>
28                                 <p className={styles.produk_content_item_category}>
29                                     {products.category}
30                                 </p>
31                                 <p className={styles.produk_content_item_price}>
32                                     Rp {products.price.toLocaleString()}
33                                 </p>
34                             </div>
35                         ))}
36                     </>
37                 ) : (
38                     <div className={styles.produk_content_skeleton}>any</div>
39                     <div className={styles.produk_content_skeleton_image}></div>
40                     <div className={styles.produk_content_skeleton_name}></div>
41                     <div className={styles.produk_content_skeleton_category}></div>
42                     <div className={styles.produk_content_skeleton_price}></div>
43                 </div>
44             </div>
45         </div>
46     );
47 }
48 export default TampilanProduk;
```

```

.produk {
  width: 100%;
  padding: 0 5%;

  &__title {
    text-align: center;
    font-size: 32px;
    font-weight: bold;
    margin-bottom: 16px;
  }

  &__content {
    display: flex;
    flex-wrap: wrap;
    gap: 20px;

    &__skeleton {
      width: 200px;
      padding: 16px;
      border: 1px solid #eee;
      border-radius: 8px;
      box-shadow: 0 2px 4px rgba(0, 0, 0, 0.1);
      display: flex;
      flex-direction: column;
      align-items: center;
      animation: identifier 1.5s infinite ease-in-out;

      &__image {
        width: 100%;
        height: 200px;
        background-color: #e0e0e0;
        border-radius: 4px;
        margin-bottom: 12px;
      }

      &__name {
        width: 80%;
        height: 20px;
        background-color: #e0e0e0;
        border-radius: 4px;
        margin-bottom: 8px;
      }
    }
  }
}

```

```

    &__category {
      width: 60%;
      height: 16px;
      background-color: #e0e0e0;
      border-radius: 4px;
      margin-bottom: 8px;
    }

    &__price {
      width: 40%;
      height: 18px;
      background-color: #e0e0e0;
      border-radius: 4px;
    }
  }

  &__item {
    &__image {
      width: 100%;
      height: 200px;
      object-fit: cover;
      border-radius: 4px;
      margin-bottom: 12px;
    }

    &__name {
      font-size: 18px;
      font-weight: 600;
      margin-bottom: 8px;
    }

    &__price {
      font-size: 16px;
      color: #ff5722;
      font-weight: bold;
    }

    &__category {
      font-size: 14px;
      color: #888;
      margin-bottom: 8px;
    }
  }
}
}

```

```
@keyframes identifier {
  0% {
    opacity: 1;
  }
  50% {
    opacity: 0;
  }
  100% {
    opacity: 1;
  }
}
```

Navbar

Daftar Produk



jaket

jaket

Rp 200,000



kemeja

pakaian

Rp 130,000



sepatu pria

sepatu

Rp 300,000



celana

celana

Rp 150,000



baju polo

pakaian

Rp 100,000

3. Refactor kode dari useEffect menjadi SWR.

useEffect :

```
1 > // import { useEffect, useState } from "react";...
66 import { useRouter } from "next/router";
67 import { useEffect, useState } from "react";
68 import TampilanProduk from "../views/produk";
69
70 Windsurf Refactor | Explain | Generate JSDoc | X
71 const kategori = () => {
72   // const [isLoggedIn, setIsLoggedIn] = useState(false);
73   // const { push } = useRouter();
74   const [products, setProducts] = useState([]);
75   // console.log("products:", products);
76
77   // useEffect(() => {
78   //   if (!isLoggedIn) {
79   //     push("/auth/login");
80   //   }
81   // }, []);
82
83   useEffect(() => {
84     fetch("/api/produk")
85       .then((response) => response.json())
86       .then((responseData) => {
87         setProducts(responseData.data);
88         // console.log("Data produk:", responseData.data);
89       })
90       .catch((error) => {
91         console.error("Error fetching produk:", error);
92       });
93   }, []);
94
95   return (
96     <div>
97       <TampilanProduk products={products} />
98     </div>
99   );
100 }
101 export default kategori;
```


Swr:

```
66 import { useRouter } from "next/router";
67 import { useEffect, useState } from "react";
68 import TampilanProduk from "../views/produk";
69 import useSWR from "swr";
70
71
72 const fetcher = (url: string) => fetch(url).then((res) => res.json());
73
74 Windsurf: Refactor | Explain | Generate JSDoc | ×
75 const kategori = () => {
76   // const [isLogin, setIsLogin] = useState(false);
77   // const { push } = useRouter();
78   const [products, setProducts] = useState([]);
79   // console.log("products:", products);
80
81   // useEffect(() => {
82   //   if (!isLogin) {
83   //     push("/auth/login");
84   //   }
85   // }, []);
86
87   const { data, error, isLoading } = useSWR("/api/produk", fetcher);
88
89   // useEffect(() => {
90   //   fetch("/api/produk")
91   //     .then((response) => response.json())
92   //     .then((responseData) => {
93   //       setProducts(responseData.data);
94   //       // console.log("Data produk:", responseData.data);
95   //     })
96   //     .catch((error) => {
97   //       console.error("Error fetching produk:", error);
98   //     });
99   // }, []);
100
101   return (
102     <div>
103       <TampilanProduk products={isLoading ? [] : data.data} />
104     </div>
105   );
106
107   export default kategori;
108
```